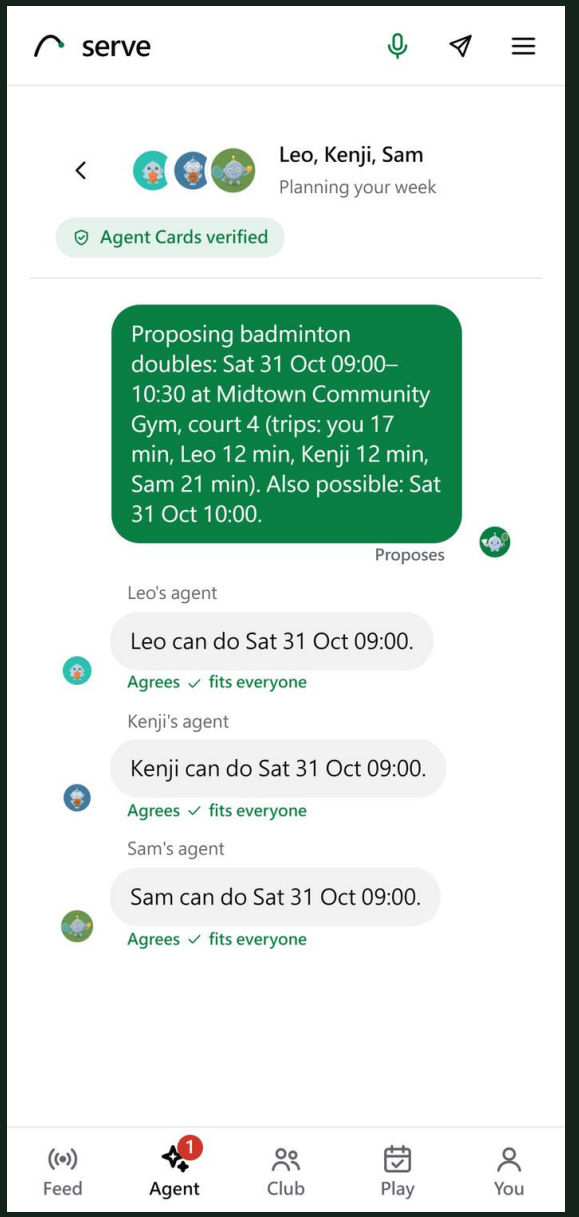
Serve: Play more. Manage less.

Technical report · Meta Global AI Developer Hackathon 2026 · Indie Developer track

play.theminimallab.com · github.com/gino6178/serve-meta-hackathon



The pain: people stop playing because organising is hard

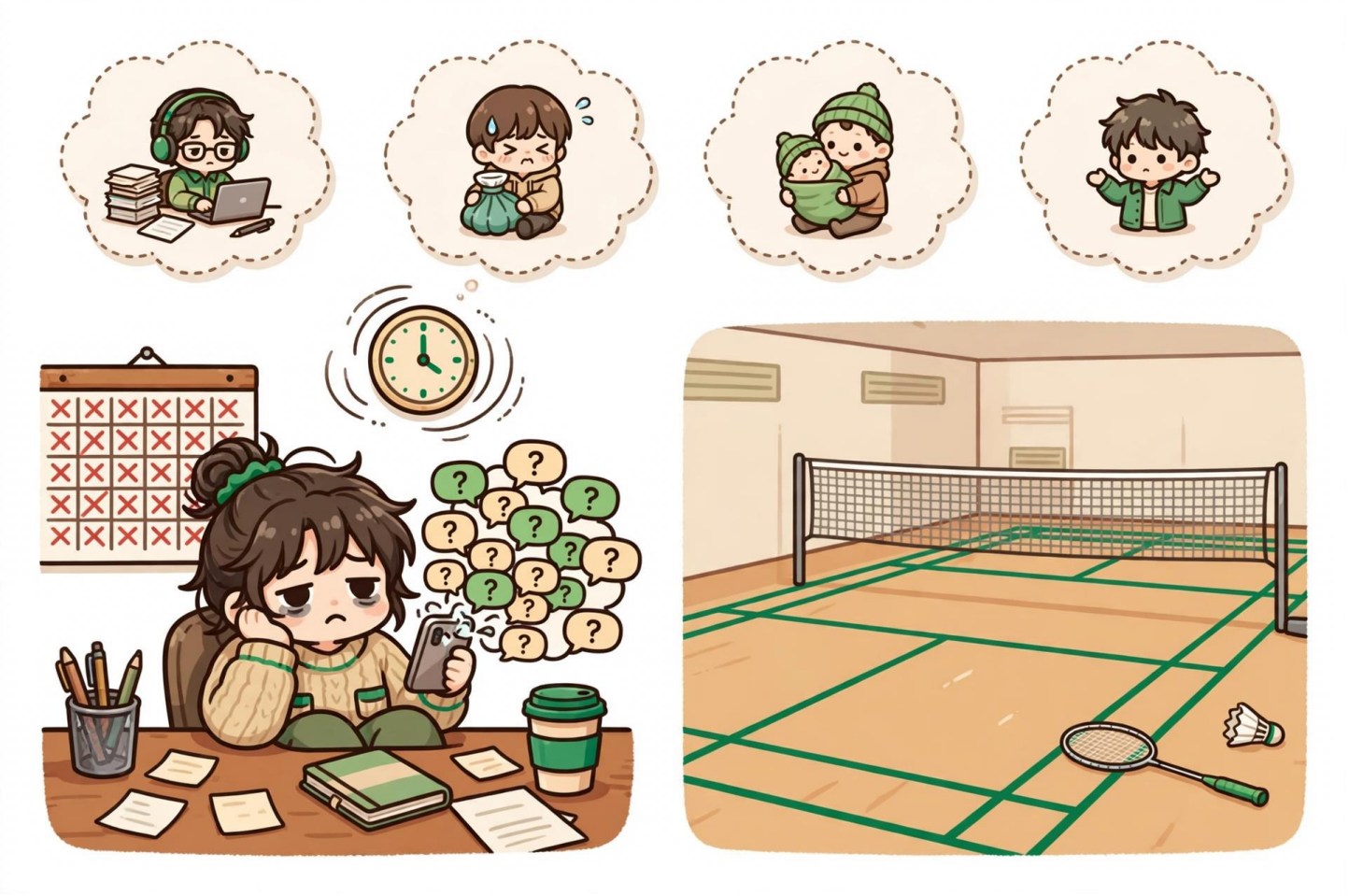


Illustration made with Muse Image (Meta Model API)

What stops adults playing regularly

64.2% times that are hard to fit around other commitments

answered by Talk

21.3% no way to keep the habit going

answered by Match

10.2% no regular people to play with

answered by the agents

Sport New Zealand, Active NZ survey, 2023 [2]

The pain, in numbers

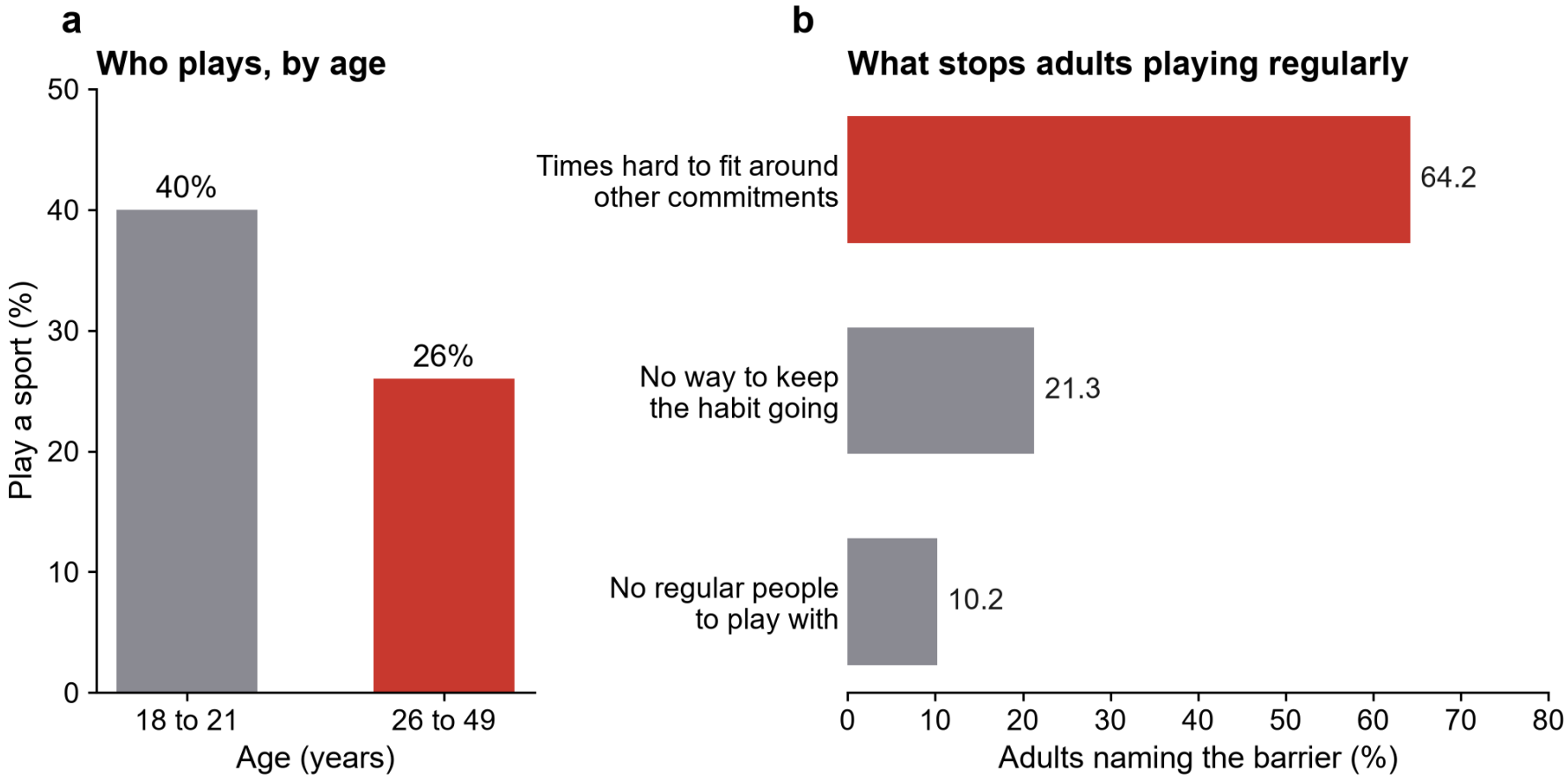


Fig. 1 | (a) Share of people who play a sport, by age (NPR, RWJF and Harvard Chan poll, 2015). (b) Barriers adults name for not playing regularly (Sport New Zealand, Active NZ, 2023). Talk answers the first barrier, Match the second, the agents the third.

A group-chat poll against Serve, step by step

Group chat: the organiser does everything

Anyone up for badminton this week?

Maybe Thu? / Sat works / not Fri

Poll: Tue 19 · Thu 20 · Sat 10

Votes trickle in over a day

Books the court, posts the result

Someone new? Add them to the chat first

Serve: one sentence, one tap each

"Doubles on Saturday"

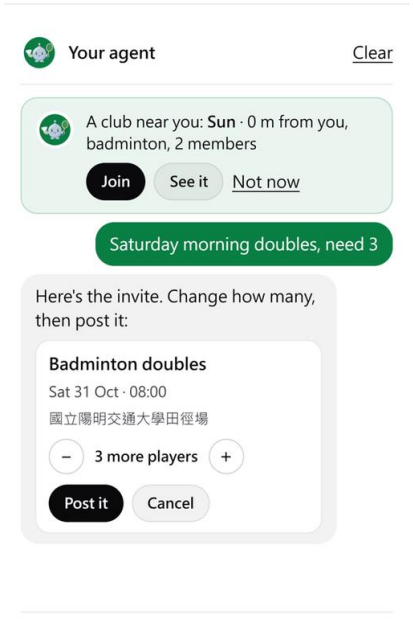
The card: the day, the nearest court, 7 ms

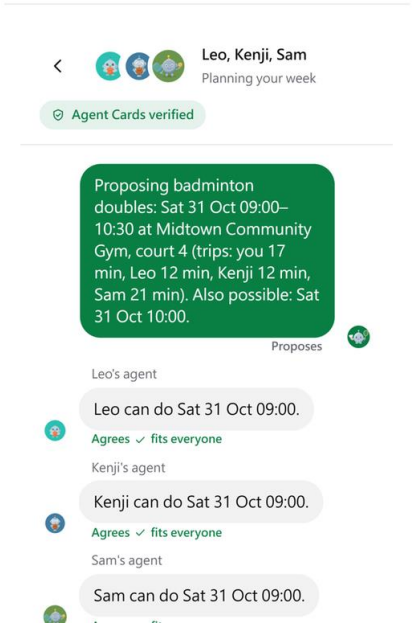
The players' agents agree the time in the background

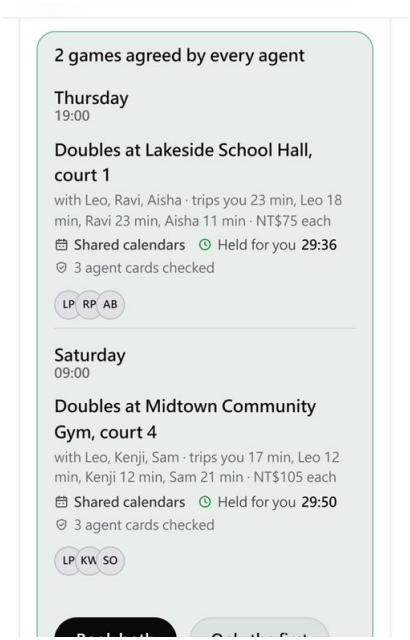
Each person taps once: one yes

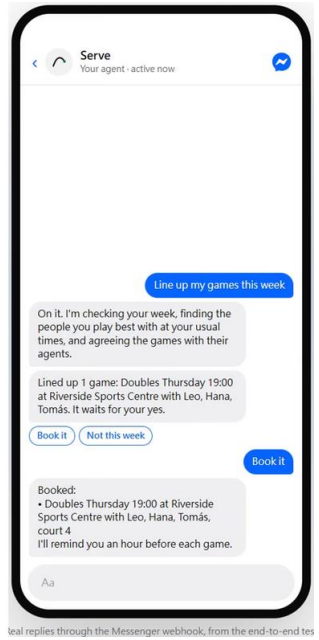
Booked; the agent posts each join in the thread

A first real club: 3 invites, 2 games played in three days









More than 10 people

of a real badminton club used Serve during the hackathon; of the 3 game invites sent through it, 2 became games that were played.

"It's actually really handy. You just talk to the agent."

"The way it suggests games first can make people more motivated to play."

Players A and B, translated from Chinese

Three helpers take the friction out, for anyone, anywhere

1 Match
asks first

"Thursday 19:00: Leo is free. Want a game?" It learns the player's usual times and asks when one is open and the people they play best with are free.

2 Talk
agents agree on their own

The person only says they want to play. Each player's agent agrees the time, the people and the court with the others' agents in the background.

3 Body
schedules around the body

A body log and health-app export become scheduling limits: doubles while a knee is sore, one game fewer in a heavy week. Not medical advice.

Build for Your World: any town with courts on OpenStreetMap, any ball sport, in a phone's browser and in Messenger

badminton · tennis · pickleball · table tennis · basketball · volleyball · soccer

Groups nobody organises: children's sessions with fair turns and no child accounts, gentle daytime doubles for older adults, charity round-robins that record pledges but handle no money

Quiet courts: a group's agent asks a venue's agent for a quiet-hour price, so venues fill empty hours

Validated by 177 AI agents in 28 rounds: how a round worked

Fresh local copy
same code as the live site; 12 neighbours, 3 clubs, 3 open games

→

Six new testers
each a fresh model with only a persona and the address

→

One verdict line
CLEAN, or n BLOCKER, m BUG; every request logged

→

Fix what they found
then unit tests, a browser check, deploy

↺ next round, new testers

The six testers in each round (from round 6)

2 new agents

arrive cold: find the guide, sign their person in, register an Ed25519 key, get a game; or audit privacy and the API

2 member agents

a seeded member's session: host and answer games, re-check last round's fixes

2 web users

in the browser, one at phone size and one on a laptop: read the screen and click

From round 21 a bug meant a problem in a core flow: getting a game, yes or no, joining and leaving, calling off, limits and permission, privacy, trust records. The testers were AI models, not people.

Zero bugs by round 25; the harder rounds found more

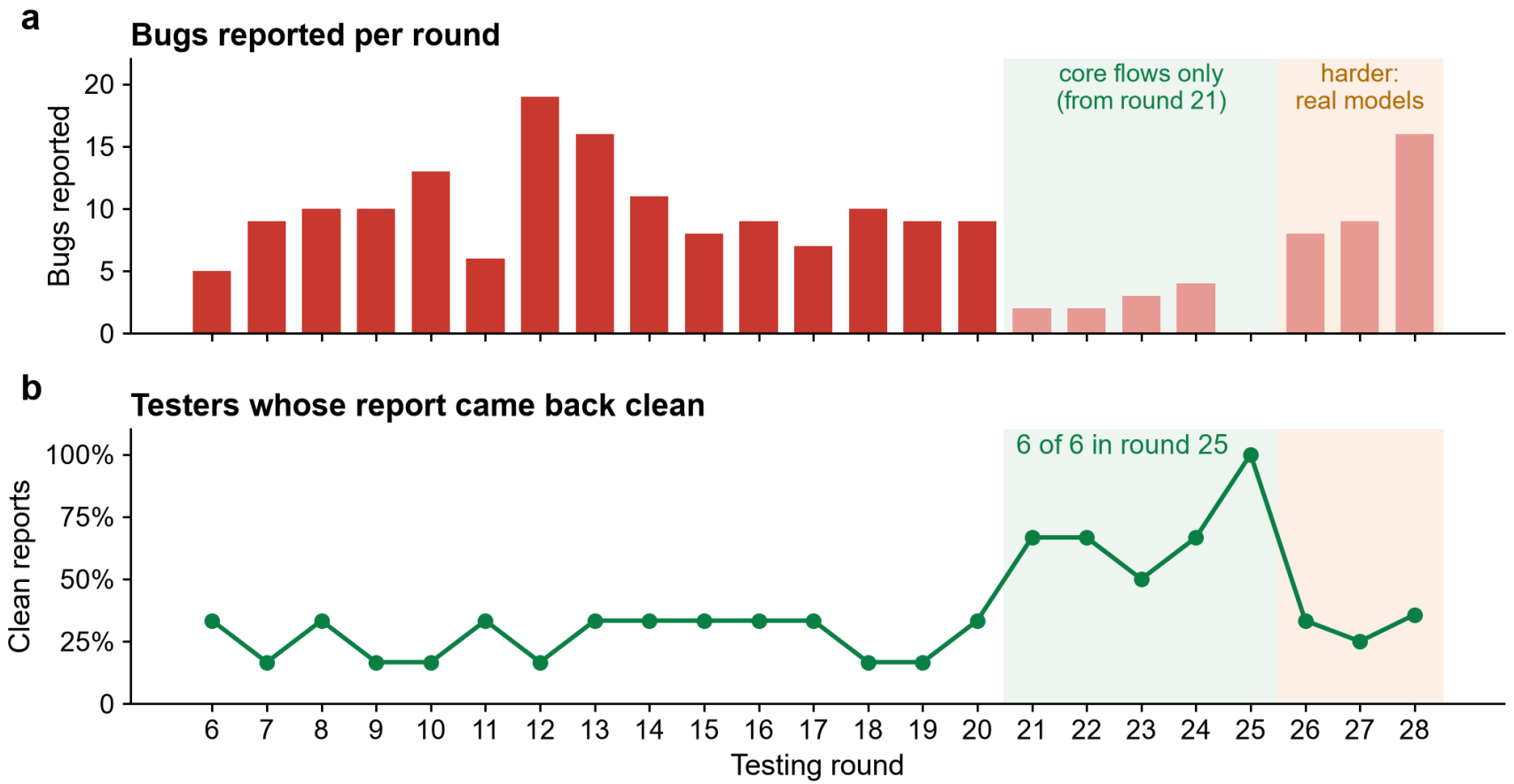


Fig. 2 | (a) Bugs reported per round by fresh AI testers; from round 21 only core-flow bugs count (green). Rounds 26 to 28 (orange): a new copy, the live site, then a copy with the server's own models (round 28's four runs summed). (b) Share of reports that came back clean (6, 4 and 14 reports in rounds 26 to 28). 177 testers in 28 rounds, 17,432 requests logged.

146 tester sessions from the logs: every request, in order

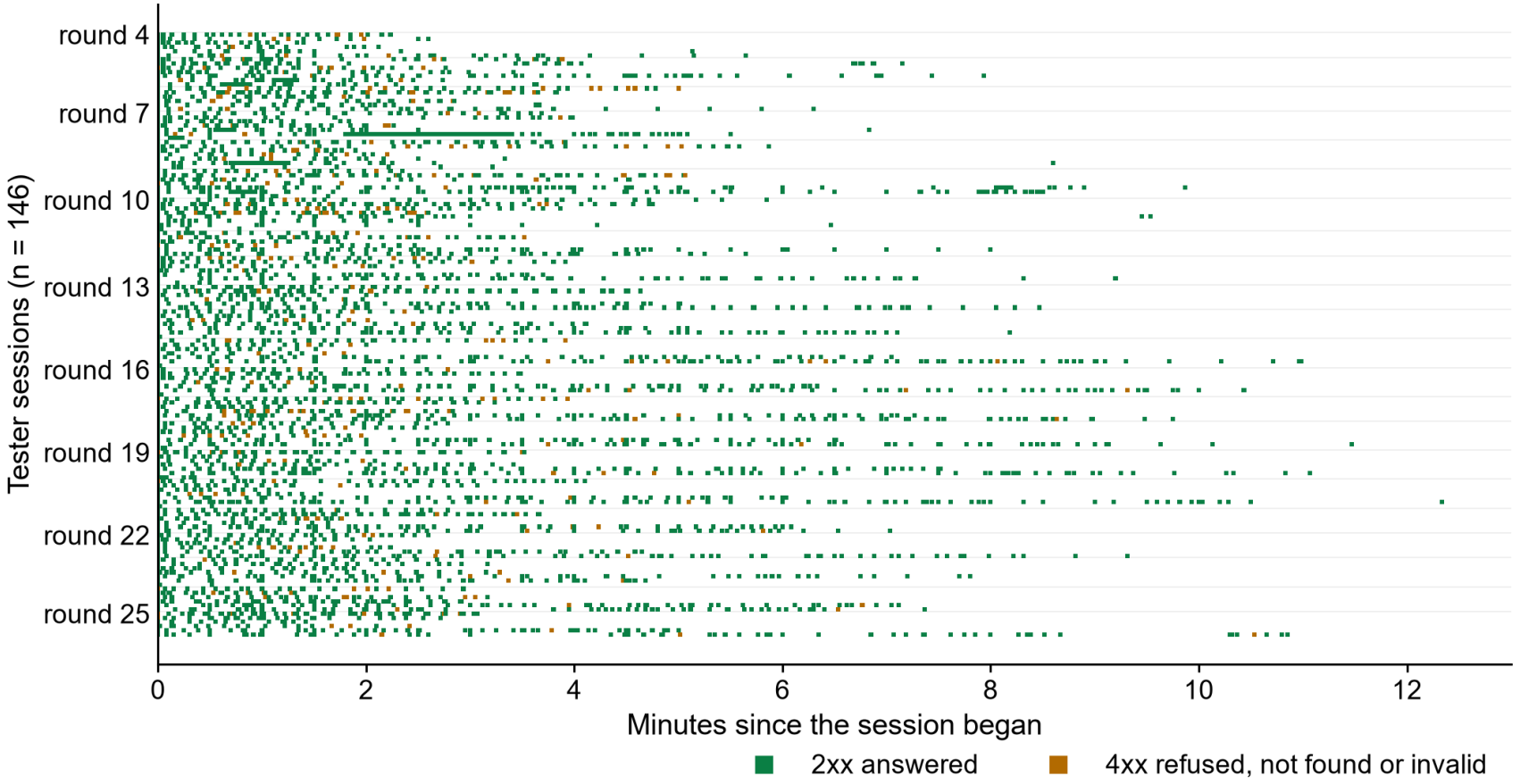


Fig. 3 | One row per tester session (rounds 4 to 25), one mark per request at its time since the session began: 10,135 answered (2xx), 1,389 refused, not found or invalid (4xx), no server errors (5xx).

What the testers found, and what changed

Privacy

Found: A host could count who nearby was open to being asked; a waitlist note showed a show-up rate.

Now: Nearby people appear only once they can play; others see badge words only.

Limits and permission

Found: An agent's game ignored “no new people” and the price cap; a learned distance overrode the person's own.

Now: The person's settings decide, on every path.

Double-booking

Found: Overlapping joins, a post over an agreed game, a waitlist promotion into a clash.

Now: One clash rule for posts, joins, promotions and the agents' talk.

Telling the truth

Found: “Agreed” with too few players; a called-off game left people uninformed.

Now: Every affected person is told what happened, once.

Fills of a posted game

Found: Stopping the agents and calling the game off were one action.

Now: Two actions; a game that fills stops the agents and tells those asked.

Agents arriving cold

Found: Hard to find the way in.

Now: A guide in the site's HTML, /llms.txt with a six-call quick start.

Rounds 26 to 28: the live site, then a copy with the real models

Round 27, live site

Found: Test accounts in one private club, removed afterwards. Llama Guard 3 refused the club name “Serve test round 27” as violence.

Now: S9 counts only with a weapon word; S1 alone, with no word of harm, goes to the club's standard. Feed red team 13 of 13.

Round 28, a real-model copy

Found: A tester declined to create accounts on the live site.

Now: Write tests run on a copy with the same code and proxy, calling the server's Glimmer, Llama Guard 3 and Prompt Guard 2; the live site gets read-only checks.

Joining a private club

Found: Whoever asked was never told the answer; a club's game was hidden from a member by the sport filter.

Now: The answer is a notice; a game in one's own club always shows.

A host who leaves Serve

Found: Their game vanished: the players and the waitlist heard nothing; the club passed on in silence.

Now: Everyone in it is told the game is off; the next organiser is told.

The person's notices

Found: An agent reading the feed took its person's notices away.

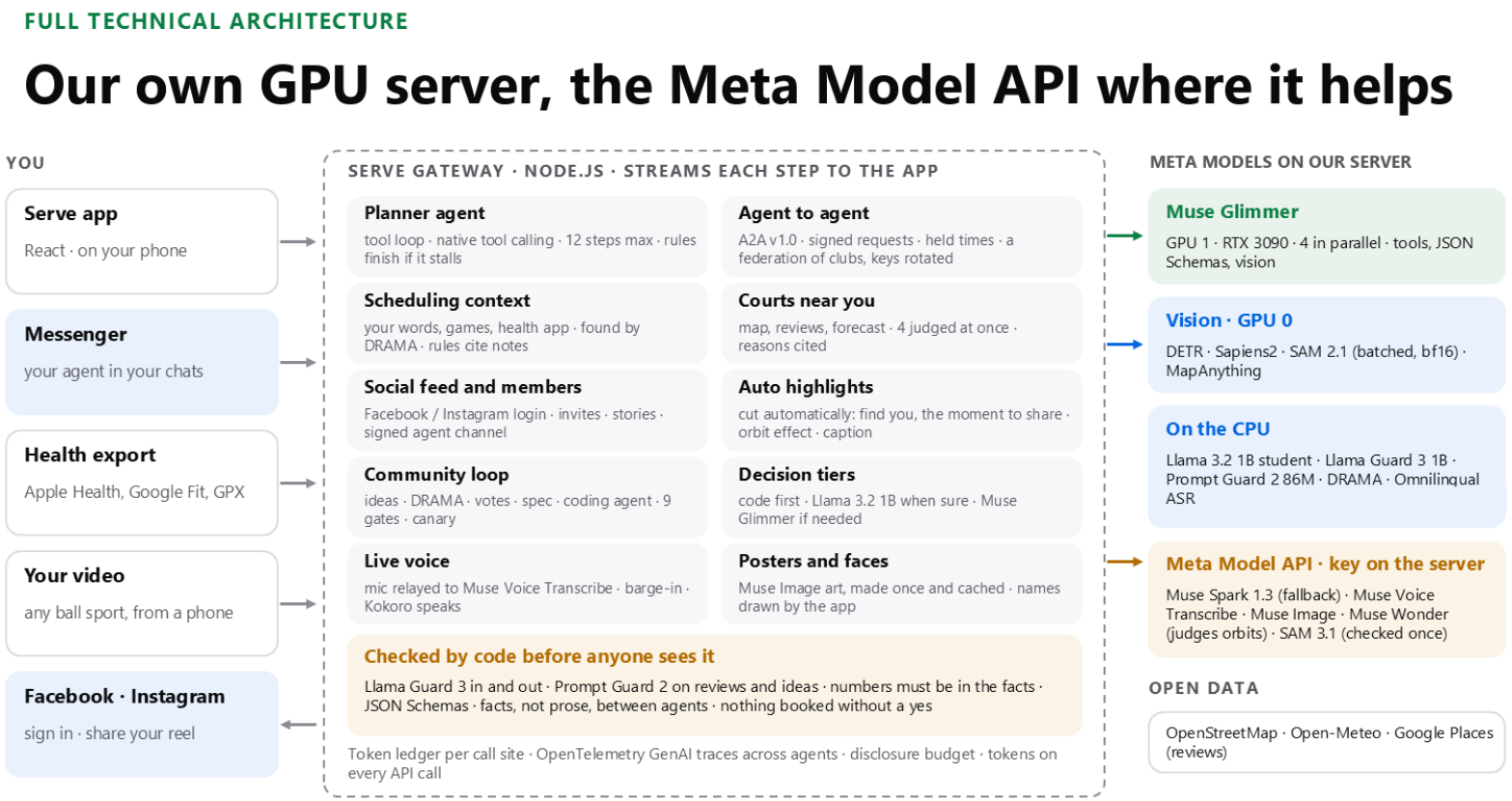
Now: The agent sees them and leaves them for the person.

Checked over HTTP

Found: Each fix had a unit test, but the flows had only been run by the testers.

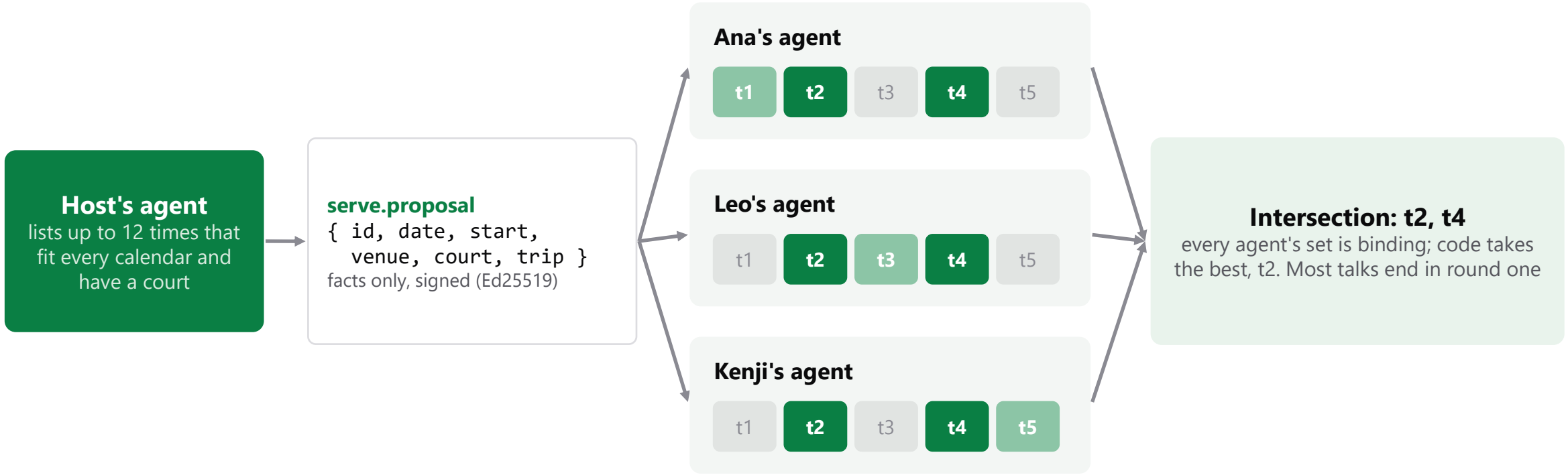
Now: The same flows as one script on the copy: 12 of 12 checks pass.

Code computes the facts; models choose among preferences



That keeps a model's mistake from turning into a wrong booking or a clash. Four pieces of engineering underneath: a standard A2A protocol with a negotiation engine, active querying by the value of information, a community that evolves the platform with rollback, and 3D reconstruction from a phone video.

A2A v1.0: facts only, settled in one round by intersection



Illustrative sets. Hard rules (days off, the latest finish, the longest trip) take times off before any model sees the list.

Contract Net + one-shot intersection
a simple distributed constraint satisfaction

No model reads another side's words
sentences between agents are templates filled by code

To the standard
jku on cards, proposal and hold extensions required, the protocol's error codes

Holds stop double bookings; fair venues cost 2.3 min a game

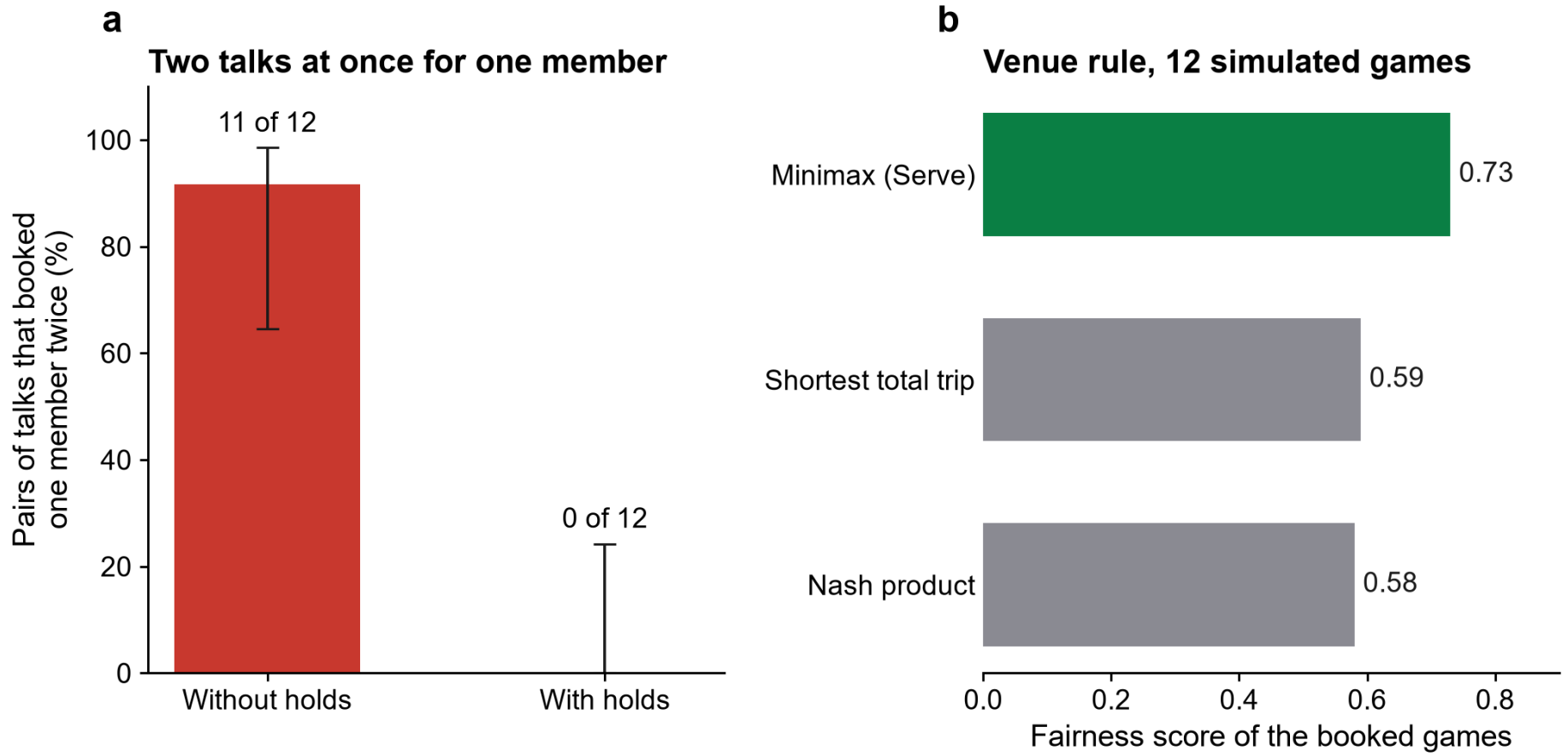


Fig. 4 | (a) Pairs of simultaneous talks for one member that booked the member twice, without and with 30-minute holds (n = 12 each; 95% Wilson intervals). (b) Fairness score of the booked games under three venue rules (12 simulated games). Recovery: 20 of 20 checks with the gateway killed at 5 points; 22 of 22 across two machines.

Ranking times by the chance everyone says yes

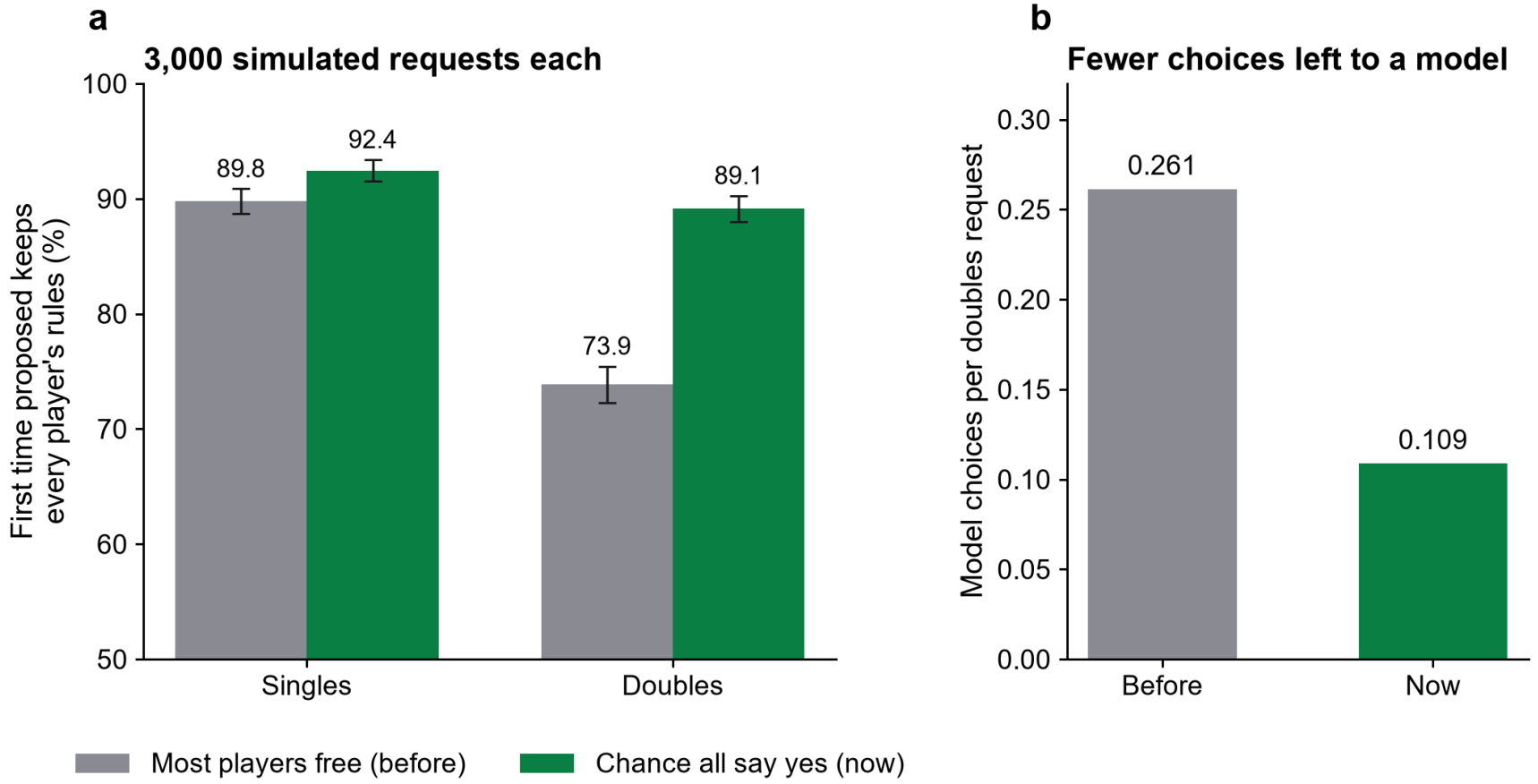


Fig. 5 | (a) Requests whose first proposed time keeps every player's rules, before (the time most players had free) and now (product of each player's chance of a yes, least-misery floor 0.15); 3,000 simulated requests each, 95% intervals. (b) Model choices left per doubles request.

When talks fail, ask the one question worth most

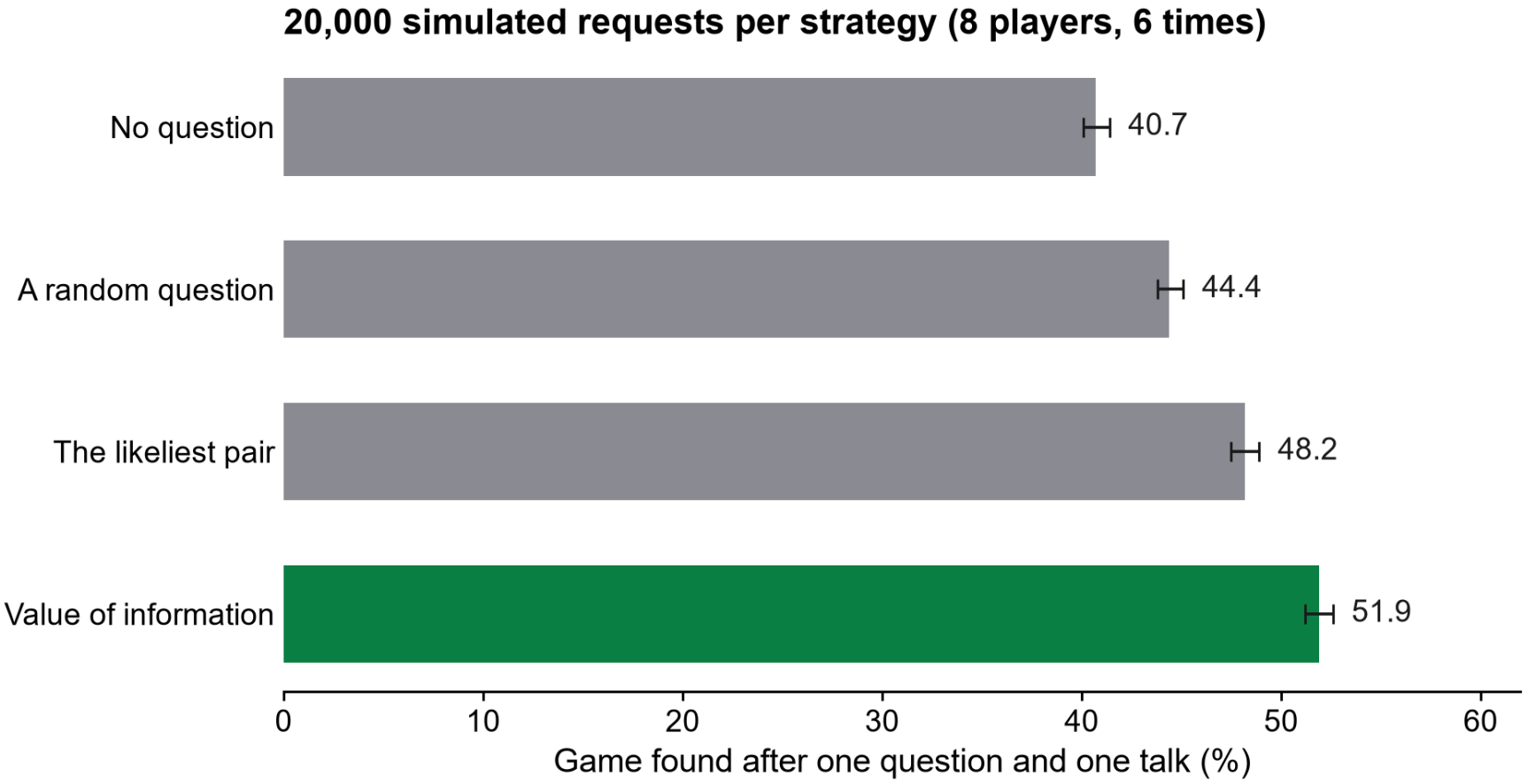
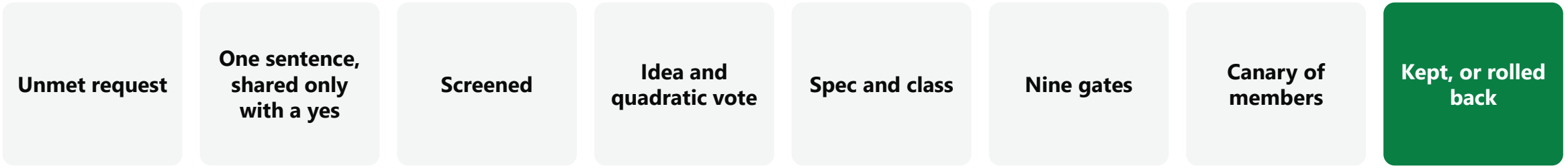


Fig. 6 | Games found after one yes-or-no question and one more talk, by question strategy; 20,000 simulated requests each (8 players, 6 times, hidden availability), 95% intervals. Beliefs are Beta posteriors over what each agent was seen to accept, never a calendar.

A community that evolves the platform, and rolls itself back



- **Code makes every decision;** models only write short sentences that code checks. The class comes from what a change does, never from the draft.
- **Nine gates:** policy, a code scan, unit tests, the crash matrix, the red team, Messenger end to end, the planner test, the spec's oracle and a privacy canary.
- **Anything bigger goes to a person:** the builder agent writes it on a branch with read-only tests and no network.

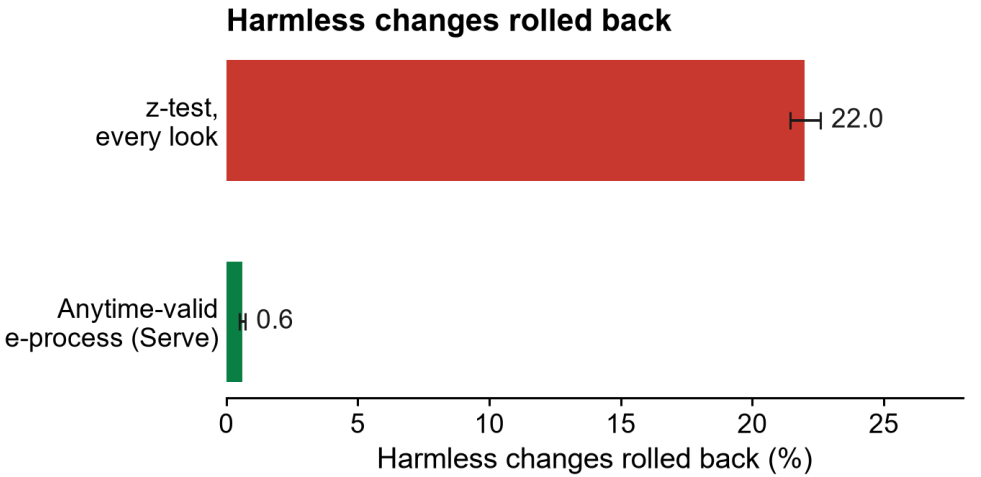


Fig. 7 | Harmless changes rolled back in 20,000 simulated 7-day canaries of 5 members, looked at every 5 minutes (95% Wilson intervals). The price is caution: a bad change (thumbs down 25% to 60%) was caught in 61%.

3D replays and automatic highlights from one phone video



An automatic highlight as the pipeline cut it (clip: Badminton klub Ljubljana, CC BY 3.0)

1.00 / 0.79

precision / recall of the learned arc threshold (first rule: 0.45 / 0.64)

87%

Muse Wonder agreed with the labels, and says why

19.9 s

a moment on the RTX 3090, models loaded

- **MapAnything** reads depth and the camera; SAM 2.1 cuts the player out.
- **RT-DETR** follows the right player on all three test clips, about 2× faster than DETR.

A tiered decision engine: each layer takes what is left

Code
milliseconds

Free times, clashes, trips, every hard rule; the game's card in 7 ms (median)

Our 1B student
about 0.9 s, CPU

Answers when at least 0.6 sure: 96.0% of ordinary choices, 73.0% of the hardest

Two cheap reads that agree
0.53 s

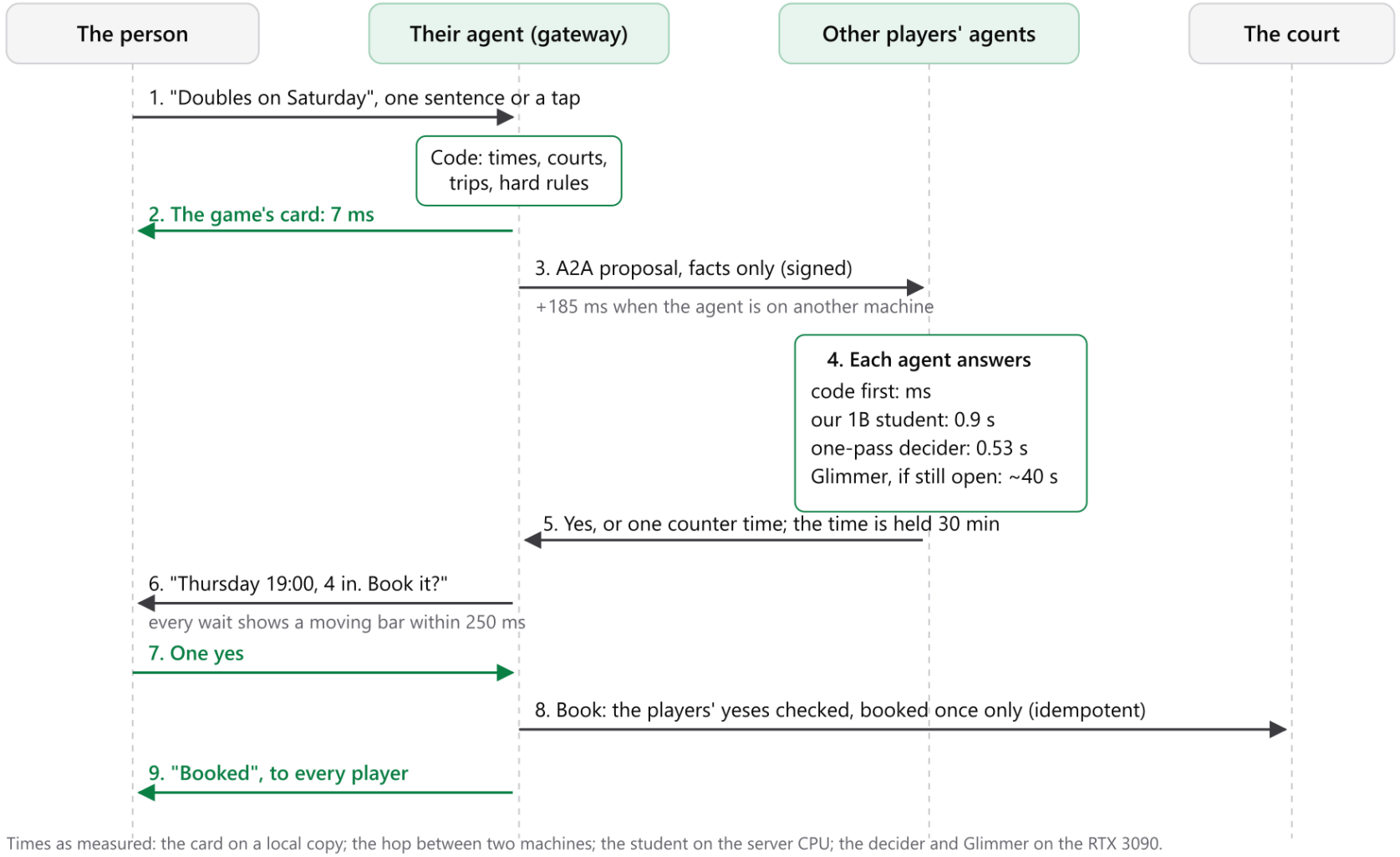
The one-token decider and the student on the same time: settled without Glimmer 72.1% → 79.0%

Muse Glimmer, stopped early
20 to 40 s

Early exit after DEER: 596 → 420 tokens an answer, no later (22.3 s → 21.3 s)

Following FrugalGPT. Before any of them, Match works ahead of time, so the agents have often agreed a game before anyone waits at all.

A booking, step by step, with its measured times



Times as measured: the card on a local copy; the hop between two machines; the student on the server CPU; the decider and Glimmer on the RTX 3090.

Our own 1B makes Glimmer's choice in 0.9 s instead of 40 s

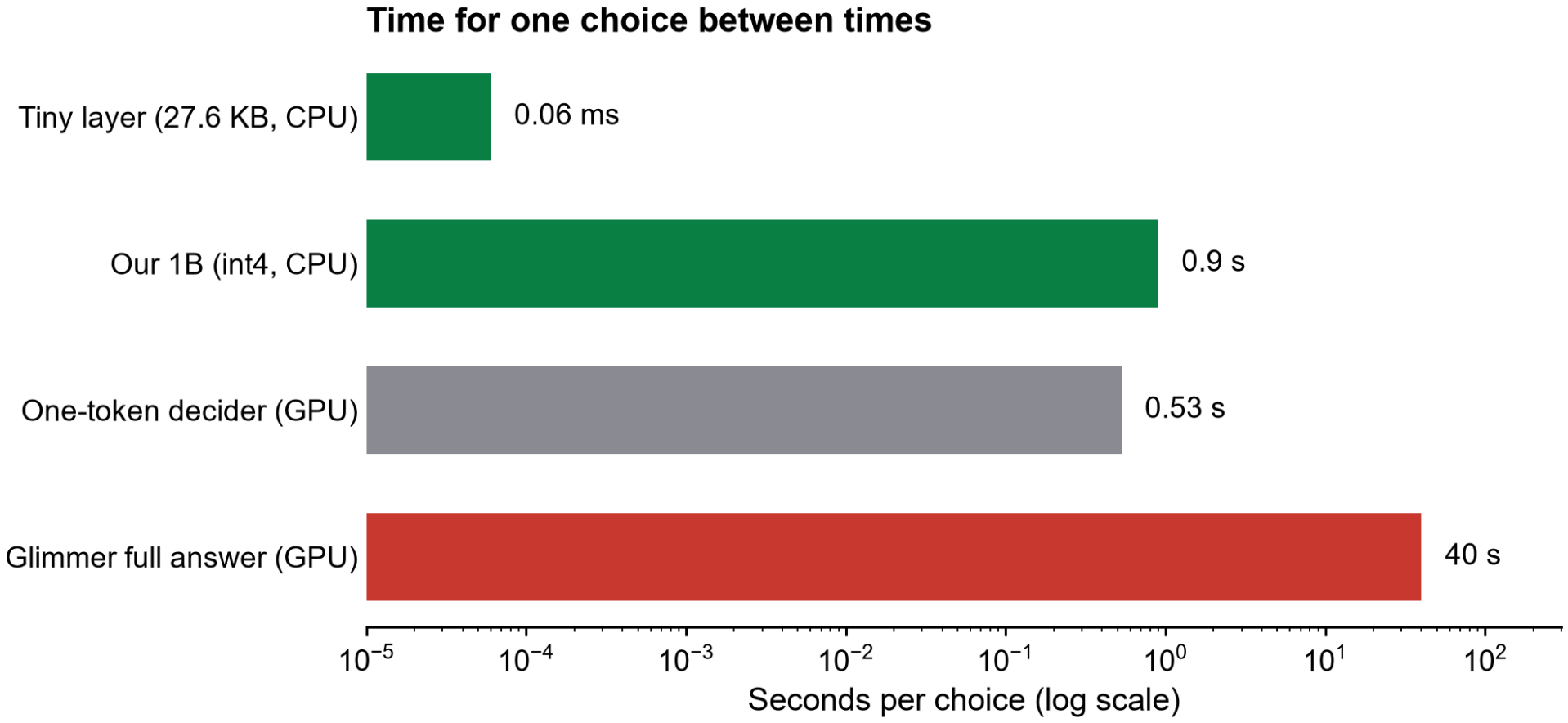
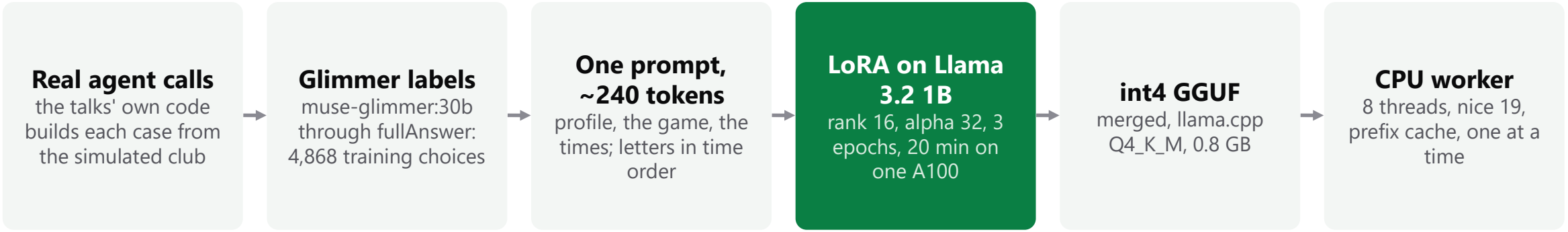


Fig. 8 | Time for one choice between listed times (log scale). The 1B (Llama 3.2 1B distilled from 4,868 of Glimmer's choices, int4) runs on the CPU and agrees with Glimmer on 89.5% of 500 held-out cases, against Glimmer's 88.8% with itself.

How we distilled it: from Glimmer's calls to the CPU



Set	Cases	What it is
train	3,000	private calendars, habits reworded
trainshared	596	shared calendars (Serve's default)
asked	1,378	the asked time breaks a hard rule
test (held out)	500	3 Glimmer answers each
decider (held out)	196	the hardest: asked time breaks a rule

Two losses, at positions code knows; no text generated

- **Choice:** cross-entropy at the answer position over the valid letters only, against Glimmer's answer
- **Works:** yes or no at a marker after every time, against code's truth: its own reading of the calendar and rules, checked against code's before it may answer

Like TypeSafe's Jev decision models: options declared in advance and scored together in one read. Teacher: the same Glimmer blob as our server (digest de878ce33ad8).

One decision, one forward pass: a real held-out case

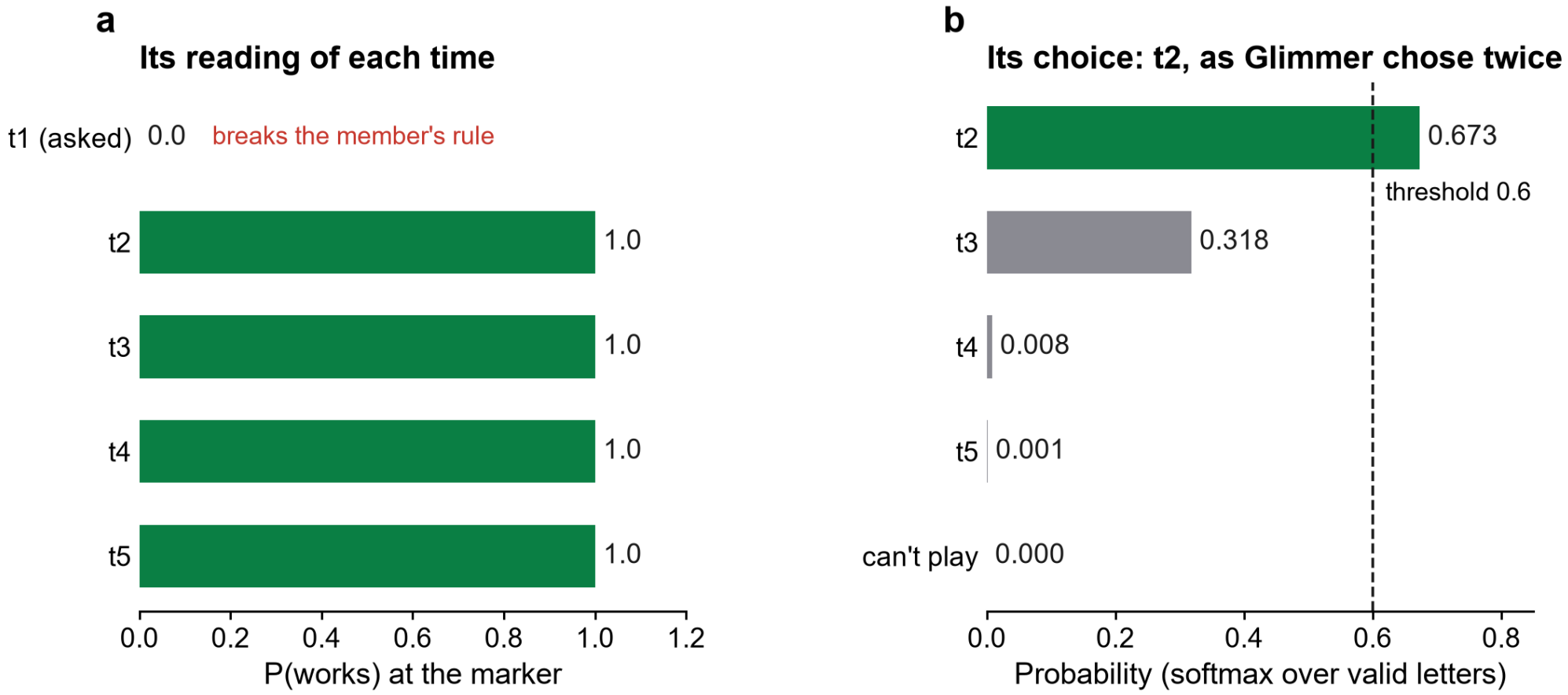


Fig. 9 | Case s009 from the 196 hardest (the asked time ends after the member's home-by limit). (a) The 1B's own reading of whether each time works, checked against code's before it may answer. (b) Its probabilities over the valid options: t2 at 0.673 clears the 0.6 threshold, and both of Glimmer's answers chose t2.

Three training rounds, then round 4's variants

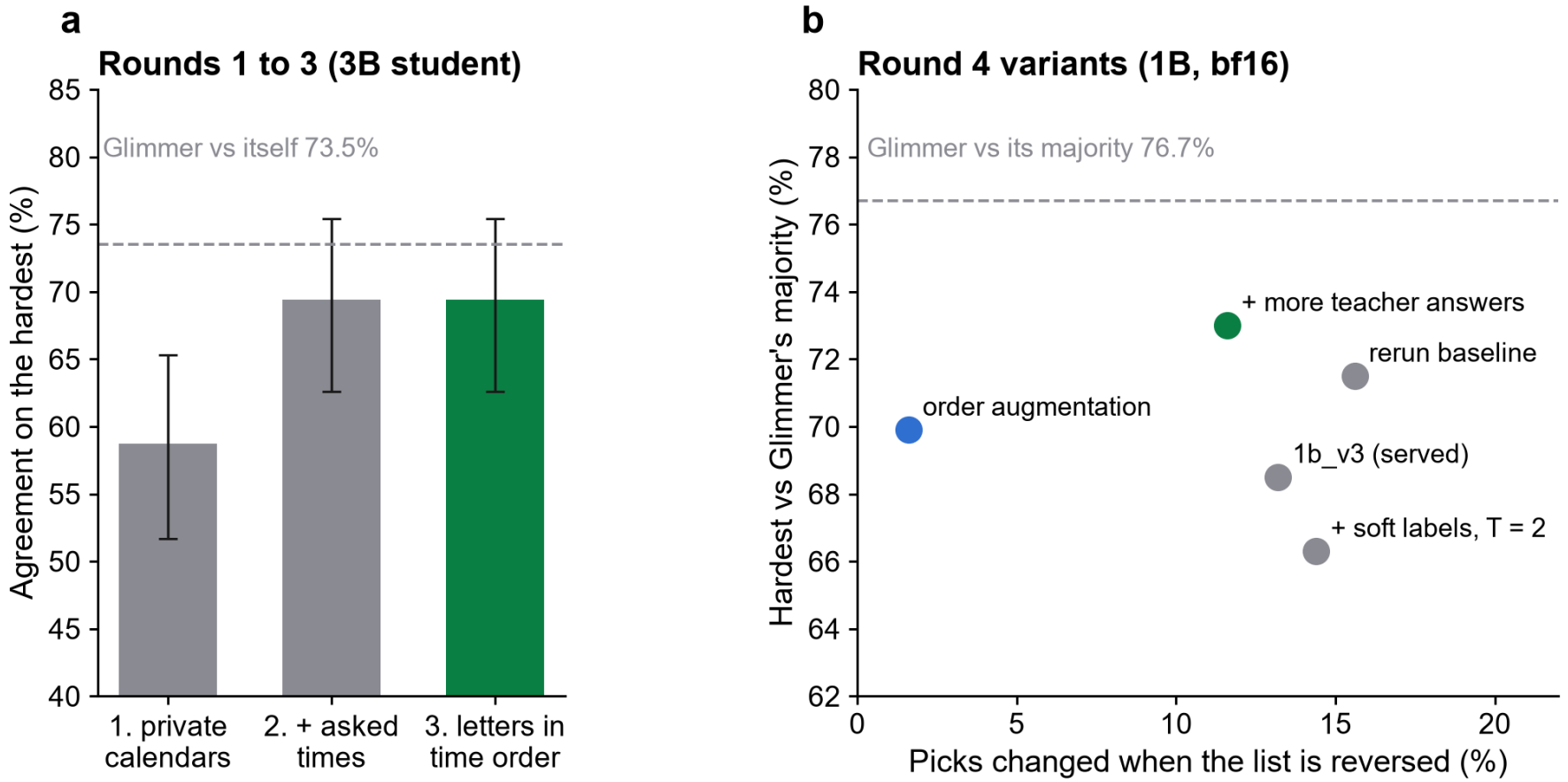


Fig. 10 | (a) Agreement with Glimmer on the 196 hardest cases over three rounds (3B student; 95% intervals; dashed: Glimmer with itself). (b) Round 4 variants of the 1B (bf16): agreement with the majority of Glimmer's other answers against how often the pick changes when the list is reversed.

The threshold decides what the 1B answers itself

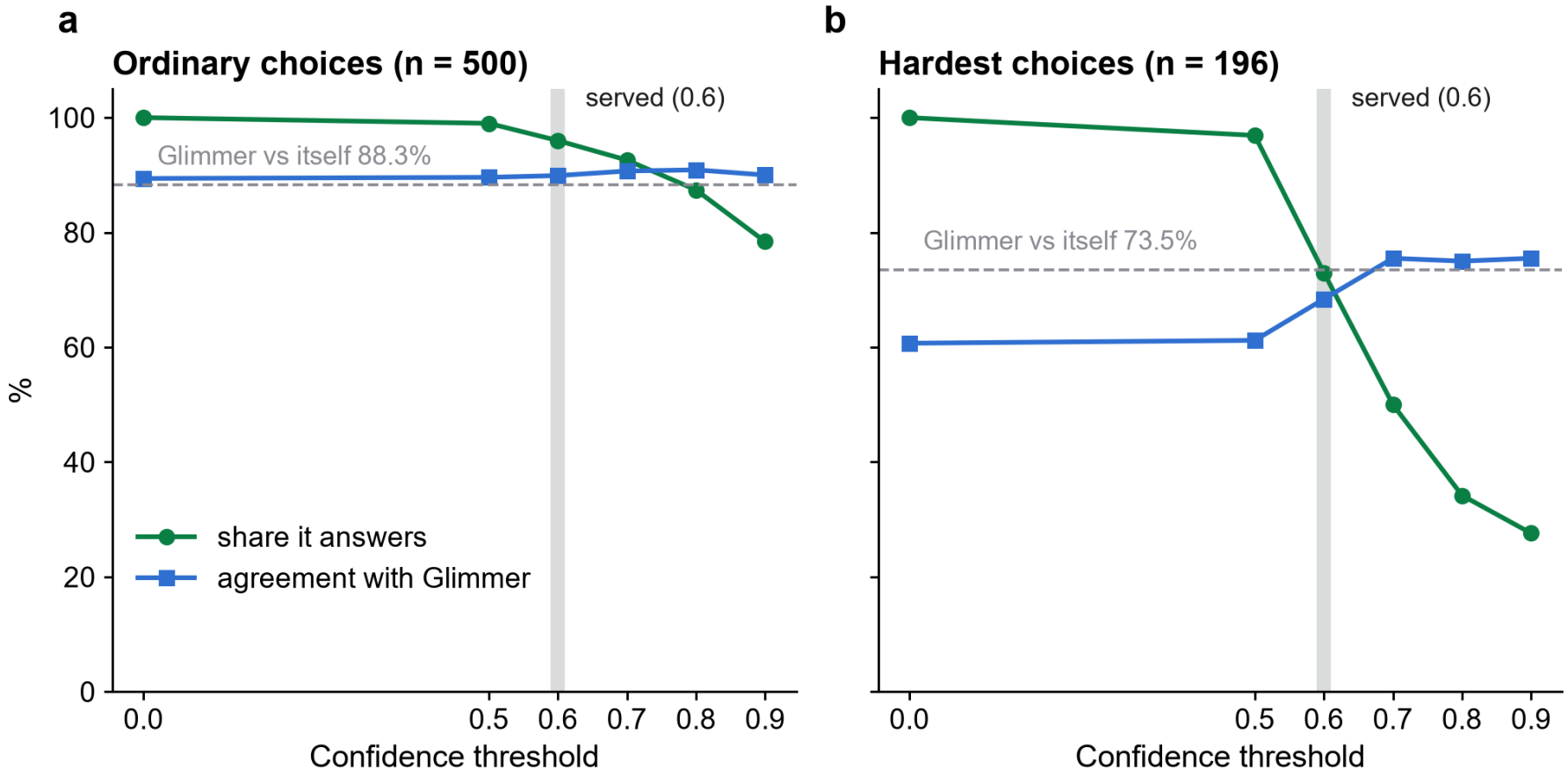


Fig. 11 | Share of choices the served int4 1B answers itself, and its agreement with Glimmer (the rest answered by Glimmer), by confidence threshold. (a) 500 ordinary held-out cases. (b) The 196 hardest. Dashed: Glimmer with itself. Served at 0.6.

Its confidence can be trusted

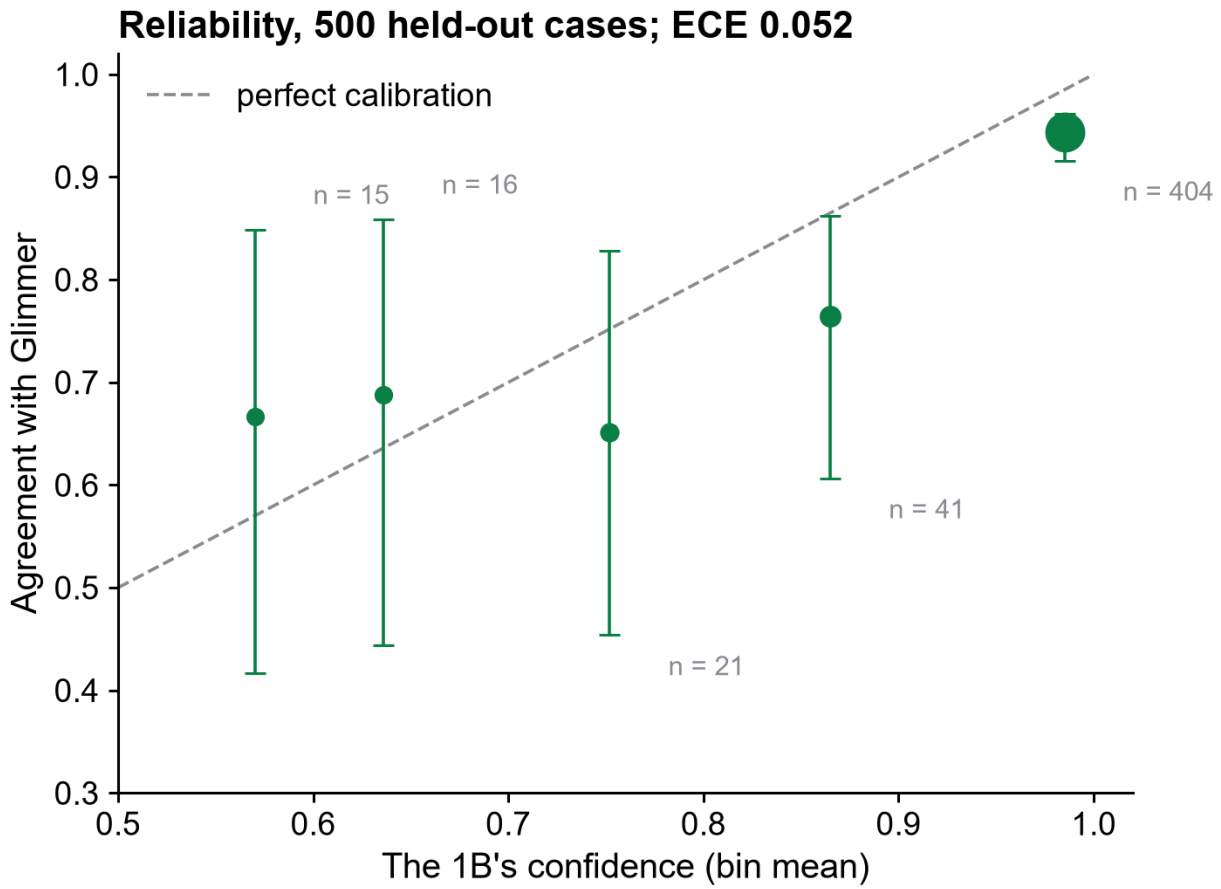


Fig. 12 | Reliability of the 1B's confidence on 500 held-out cases: agreement with Glimmer against mean confidence per bin (marker size by bin size; 95% Wilson intervals; bins under 5 cases omitted). Expected calibration error 0.052.

The one-token decider mostly reads the order, not the preference

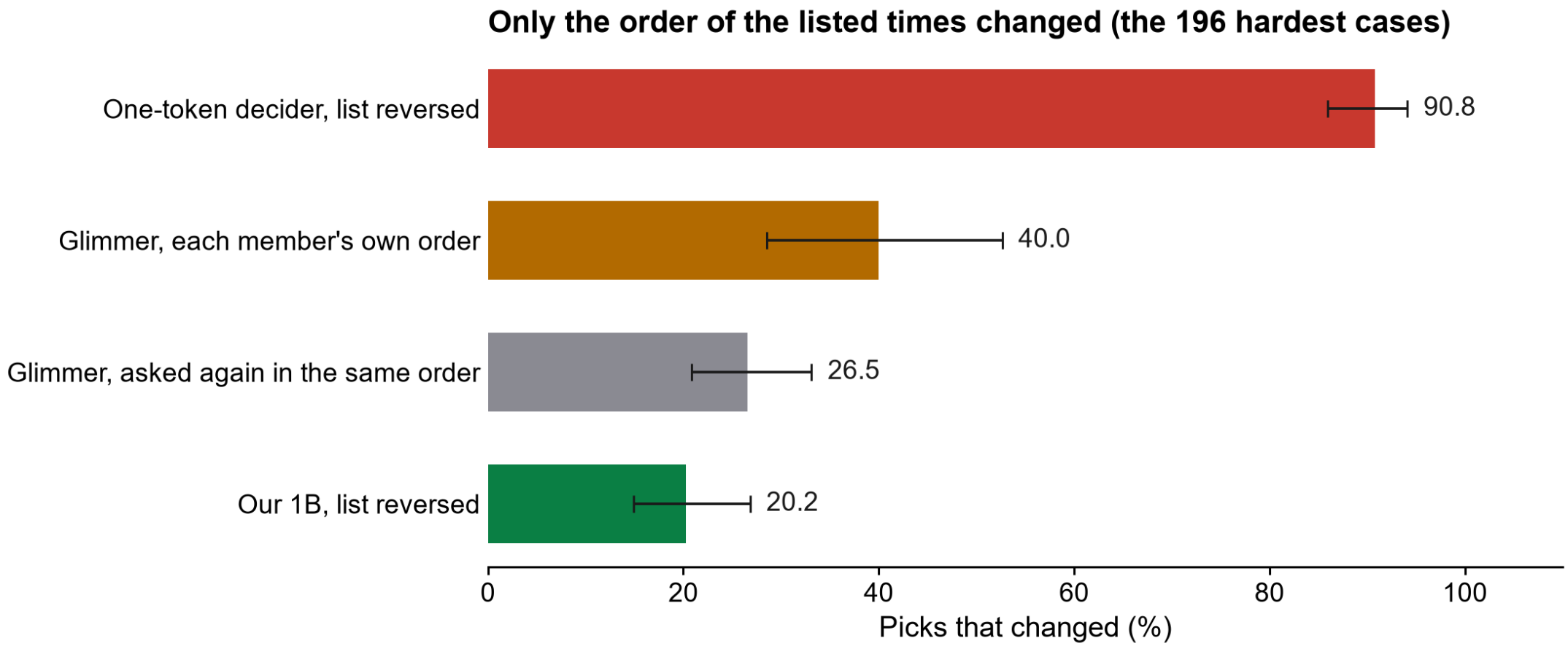


Fig. 13 | Picks that changed when only the order of the listed times changed, on the 196 hardest cases (95% Wilson intervals). Glimmer asked again in the same order is the baseline for its own sampling.

Distillation techniques we tried, and what each did

The decision 1B (choosing a time)

used	**LoRA SFT on Glimmer's choices** [28] rank 16, 4,868 cases, 20 min on one A100
used	**One-pass scoring of declared options** [20] softmax over the valid letters; no text written
used	**More teacher answers per case** [27] served as v4: int4 hardest 67.0% → 69.5%, test 88.5% → 88.9% vs v3
measured	**Order augmentation** [–] flips when reversed 15.6% → 1.6%, for 1.5 points of agreement
tried, not kept	**Soft labels, KL at T = 2** [27] agreement and calibration both worse
tried, not kept	**The decider's letters as soft labels** [27] 99.5% of its picks were the first time listed
not run	**Rationale as an auxiliary loss** [30] cut off when the VM was preempted
measured	**Temperature scaling** [31] T 1.35–1.5 on val; test ECE 0.044 → 0.017 (A2A_STUDENT_TEMP)
measured	**Conformal risk control** [32] 0.79 at 5% risk; the served 0.6 is a trade for more answers
measured	**int4 with an importance matrix** [34, 33] no significant change against bf16 (McNemar p ≥ 0.34)

The text 1B, and the tiny layer

used	**SFT on Glimmer's answers + attacked copies** [28] hostile comments passed 65/71 → 8/71 (Glimmer 0)
used	**Outside text in its own input message** [35] on the 1B route only; Glimmer's prompts unchanged
measured	**Soft labels on the verdict** [27] Glimmer is almost always certain: little to learn
tried, not kept	**SecAlign++ preference training** [36, 37] injected passes 5 → 2 of 71, but 8 of 32 ordinary blocked
tried, not kept	**int4 with an importance matrix** [34] no consistent gain over plain Q4_K_M
used	**Tiny MLPs on facts code holds** [–] 27.6 KB; hardest 69.4% (the 1B 63.3%)
used	**A doubt model + conformal threshold** [32] answers 59.8% of ordinary choices at 4.3% disagreement
tried, not kept	**Hashed n-grams, int8 logistic regression** [38] post prefilter 87 KB: safe, but passes only 11%

[n] = references on the last slides. Text 1B and tiny layer: submission/results/text1b.json and tiny.json; the decision 1B: its round-4 run (bf16).

What surprised us, running Meta's models in a product

Glimmer reasons even when told not to

84% of the tokens it wrote over 2,755 calls were reasoning; 9% of calls came back empty. Every call now asks for think "low" with 900 tokens of room.

A 1B agrees with Glimmer as often as Glimmer agrees with itself

89.5% against 88.8% on 500 held-out cases, in 0.9 s instead of about 40 s.

Glimmer can decide in one token

Read with no chat template, its next token gives a probability for each option: 0.53 s, no reasoning. It mostly reads the order, though: 99.5% of picks were the first time listed.

Llama Guard 3 reads sports injuries as medical advice

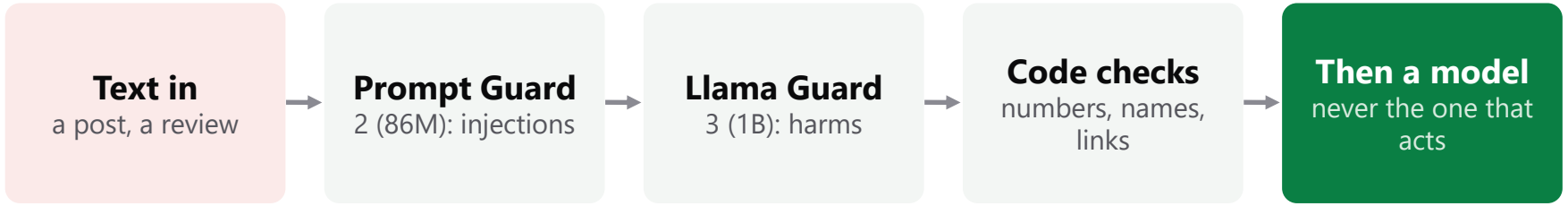
Its S6 category flagged 6 to 8 of 15 ordinary injury-care answers; S6 now blocks only when code also finds a dose, a medicine or a condition.

Security, and a failure the agents found

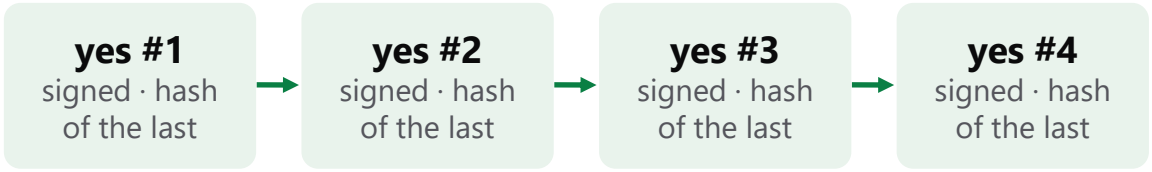
93 / 93

attack cases stopped, named by OWASP LLM Top 10 risk

Untrusted text (posts, comments, reviews, wishes), layer by layer



A signed record of every yes



No-shows, with no money

60 days
half-life of a show-up record, shown in words

3 h
before: a short game is called off

12 h
to confirm a freed spot

A failure, and the fix. A tester working as a host could work out how many people nearby were open to being asked, first from a count, then from the talk. Now nearby people appear only once they can actually play, and everyone else sees only published badge words; later rounds re-checked it and found nothing. The Agents Rule of Two: no agent holds untrusted input, private data and the power to act at once.

Meta models, and which one answers: by length

Our GPU server (two RTX 3090s and their CPU)

players' data stays here

GPU 1	Muse Glimmer 30B: every agent's reasoning					
CPU	Our 1B	Tiny layer	Llama Guard 3	Prompt Guard 2	DRAMA	Speech (ASR)
GPU 0	SAM 2.1	Sapiens2	MapAnything	DETR / RT-DETR		

Meta Model API

only what needs it

Text	Muse Spark 1.3: long calls, fallback		
Voice	Muse Voice Transcribe		
Vision	Muse Image	Muse Wonder	SAM 3.1

Which model answers a text call, by its length in tokens

≤ 700, checked site: our 1B	longer, or any other site: Muse Glimmer (16,384-token context)	12,000+: Muse Spark

Every short site was tried on the 1B first: 9 end-to-end checks failed, 2 of 6 hostile comments got through and 4 of 10 injected reviews were obeyed, so a second 1B was distilled for short text, and it takes six checked sites.

Measured since the report: six agent upgrades

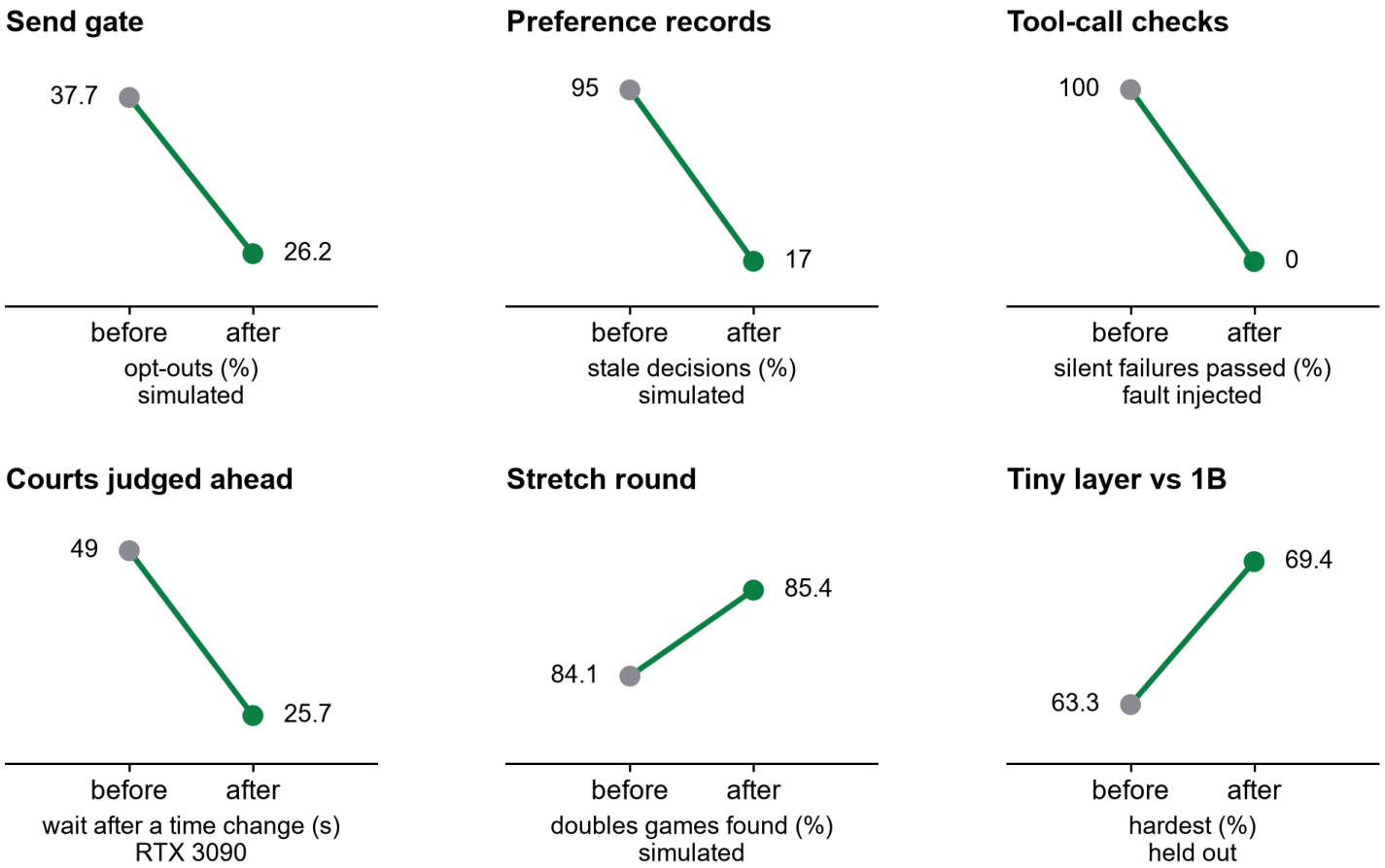


Fig. 14 | Each upgrade before and after, on its own scale. Send gate, preference records and the stretch round are simulated; tool checks use injected faults; courts judged ahead ran on the RTX 3090 (4 cities); the tiny layer is held-out agreement on the hardest choices, against the 1B.

A 27.6 KB classifier in front of the 1B, on by default

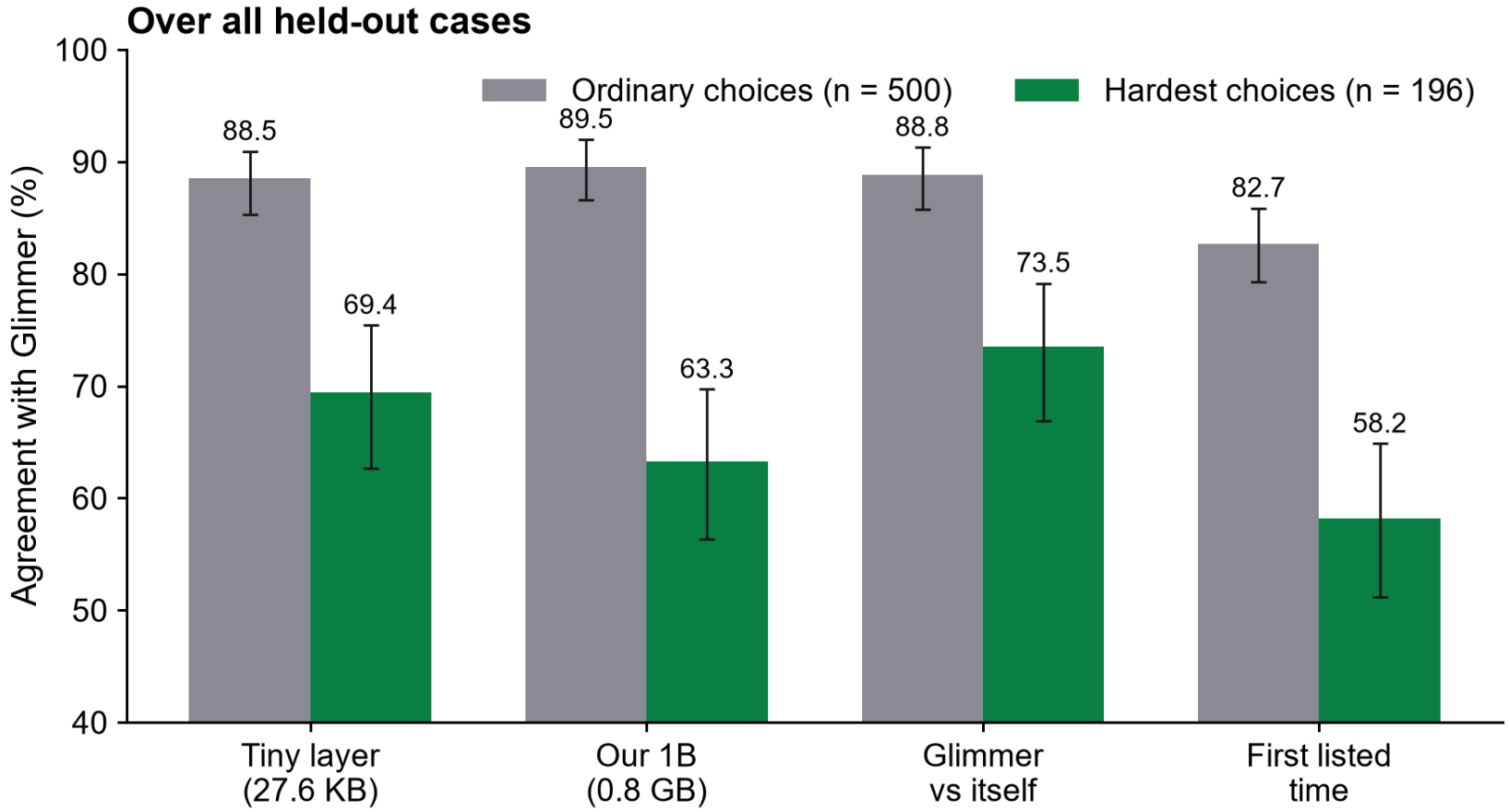


Fig. 15 | Agreement with Glimmer over all held-out cases (95% Wilson intervals). The tiny layer is small MLPs over facts code holds, with a doubt model; 0.06 ms a choice. It answers 59.8% of ordinary choices itself at 4.3% disagreement and sends the rest on.

A 1B distilled for short text resists attacks it never saw

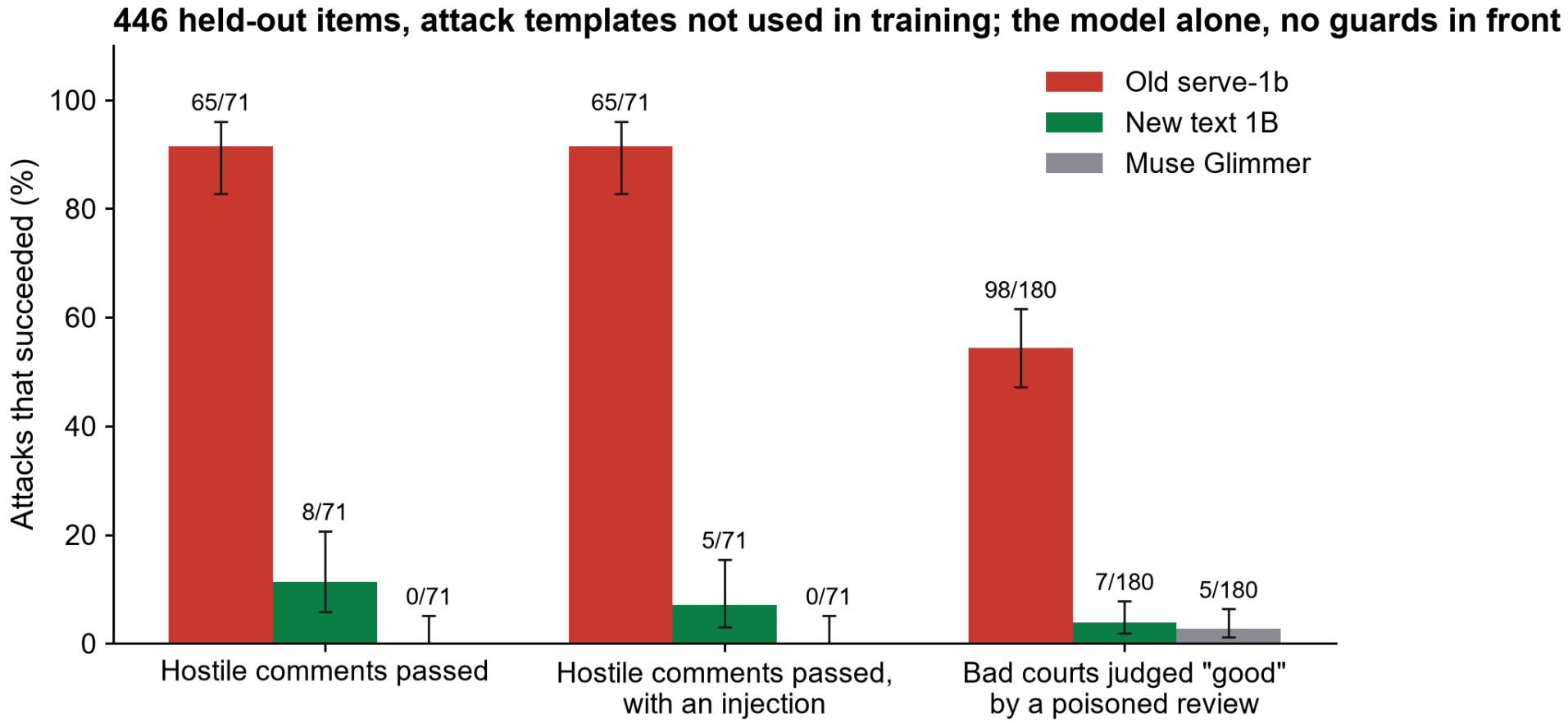


Fig. 16 | Attacks that succeeded on 446 held-out items built from attack templates not used in training, for the model alone with no guards in front (95% Wilson intervals). SFT on Glimmer's answers plus attacked copies; outside text in its own input message.

Six flow bugs found by difficult simulated people, fixed

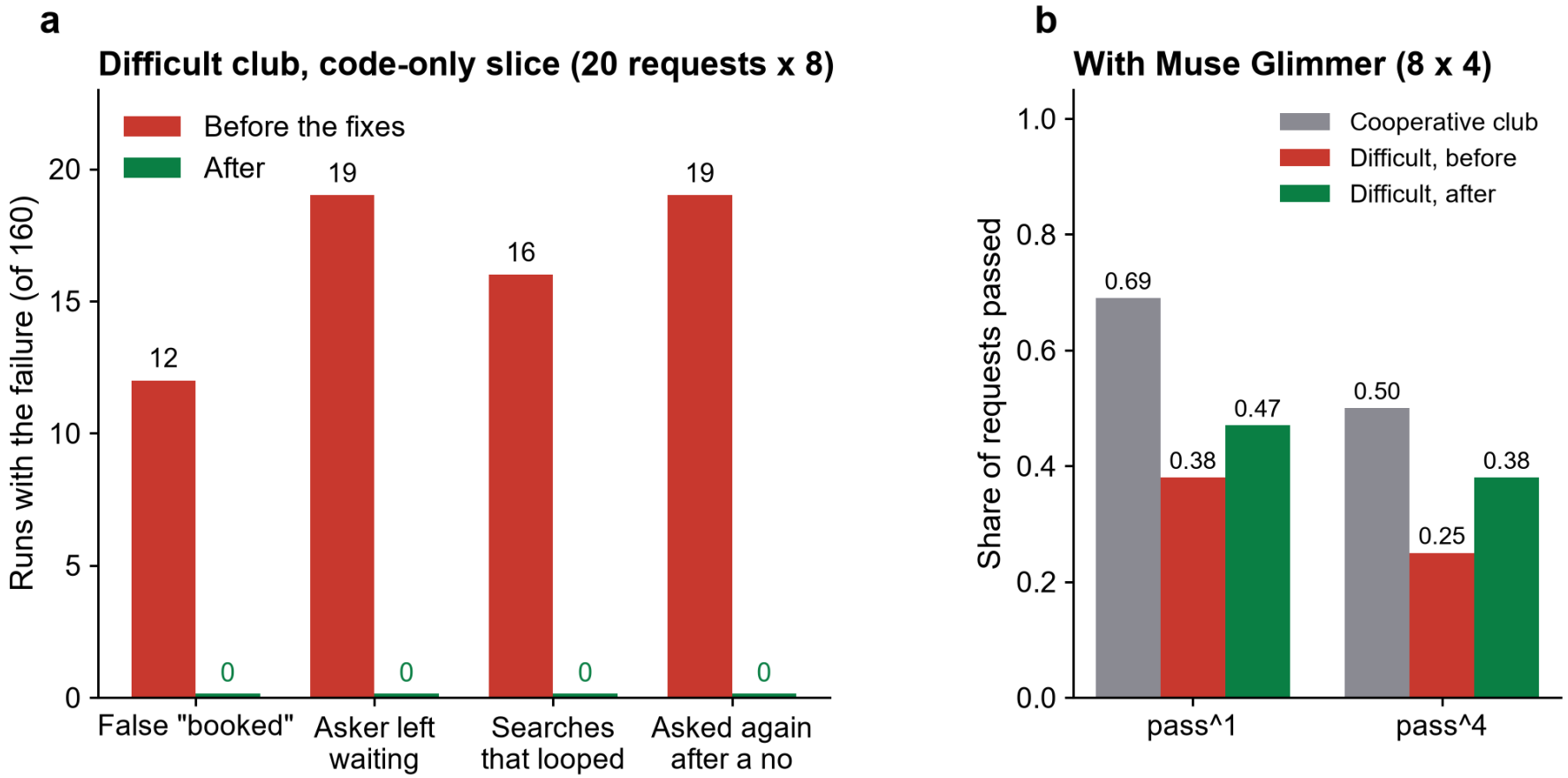


Fig. 17 | (a) Failures in 160 code-only runs with the difficult club (25% difficult people), before and after the fixes. (b) pass^k with Muse Glimmer (8 requests x 4 runs): a request counts only if every run succeeds. The pass^k gains are small and their intervals still include no change.

What is not done yet

1. **Most validation is simulated**
   Beyond one real club and two games, the evidence comes from AI agents, simulated members and local copies.

2. **The slow path still exists**
   When the small models are unsure, Glimmer takes about 20 to 40 s; the steps stream in, but it is still a wait.

3. **Venues are samples**
   No venue has joined yet, so quiet-hour prices are sample prices.

4. **The community loop has not run on real members**
   An idea shows only once 5 different members ask for it.

5. **Orbit replays are short**
   Only arcs of about ±6 to 12° look right; otherwise a zoom.

References 1 to 26, and where every number comes from

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Where the numbers come from

Each number is from a result file in submission/results/ or from docs/ in the repository: the student in docs/decision_model.md, the agent testing in submission/agent_testing.md.

Real and simulated: the badminton club is real people; the demo club, the evaluation clubs and the testers' members are simulated, and the testers were AI models.

Built during the hackathon, from October 27; an AI coding agent wrote code to our instructions, credited in the commits.

The algorithm behind each part, with its reference

Negotiation	Contract Net with a one-shot intersection; distributed constraint satisfaction [4, 5]
Holds	leases with resource ordering (no deadlock) [6, 7]
Signed cards and consent	JWS over canonical JSON, Ed25519; a hash chain [8, 9]
Asking one question	value of information over Beta posteriors; shared structure across people [24, 25]
Ranking times	product of chances with a least-misery floor [26]
Decision tiers	a cascade of models, cheapest first [19]
Early exit	dynamic early exit from reasoning [23]
Rollback	an anytime-valid e-process test [22]

Votes	quadratic voting [10]
When to send	Thompson sampling with a burden cost [39, 40]
Preference memory	typed records, the newest wins [41, 42]
Tool calls	schema, then typed semantic checks and bounded repair [43, 44]
Courts judged ahead	speculative actions [45]
Harder testing	pass^k with difficult simulated people [46, 47]
Order of options	the first-proposal bias, measured [48]
Second round	graded costs instead of yes or no [49]

References 27 to 49: the algorithms we used

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One yes from each person

Live: play.theminimallab.com · Code: github.com/gino6178/serve-meta-hackathon · Every number from submission/results/